K.S.RANGASAMY COLLEGE OF TECHNOLOGY, TIRUCHENGODE - 637 215
(Autonomous)
DEPARTMENT OF INFORMATION TECHNOLOGY

Academic Year : 2025 – 26 Year / Sem : IV / VIII

60 IT 8P1 Project Work - Phase II - Review 1 Report

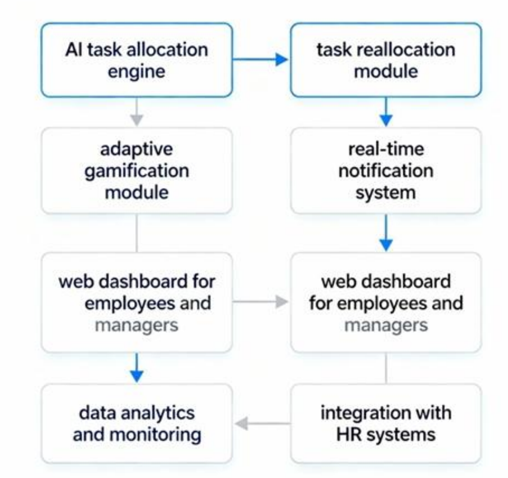
Title: AI-Based Task Reallocation and Employee Productivity System

Batch no: 29

Objective

- The primary objective of the AI-Based Task Reallocation and Employee Productivity System is to enhance organizational efficiency by ensuring fair and balanced task allocation among employees.
- The system aims to automate task assignment using real-time data on skills, availability, and workload, reducing managerial effort. It seeks to maintain continuity by dynamically reallocating tasks during employee leave or absence.
- Additionally, it focuses on motivating employees through adaptive gamification, promoting accountability and engagement.
- By providing analytics-driven insights into performance and workload trends, the system enables informed decision-making, fostering a transparent, equitable, and high-performing workplace environment.

System Architecture



Hardware Requirements

Processor Type : AMD RYZEN 7

RAM :16 GB RAM

Hard disk : 1 TB

Software Requirements

Front End : HTML, CSS

Backend : JAVA

Framework : Spring

Module description:

- **Authentication Module** – Provides secure login for admin and employees with role-based access control.
- **Employee Data Modeling Module** – Stores employee skills, availability, and workload for intelligent decision making.
- **Intelligent Task Allocation Module** – Uses a rule-based AI algorithm to assign tasks based on suitability scores.
- **Dynamic Task Reallocation Module** – Automatically reassigns tasks when an employee is absent or unavailable.
- **Gamification Module** – Motivates employees using points, badges, and leaderboard rankings.
- **Performance Analytics Module** – Displays workload trends and task performance through interactive dashboards.

Algorithm Used:

Algorithm 1: Task Allocation & Dynamic Reallocation Algorithm

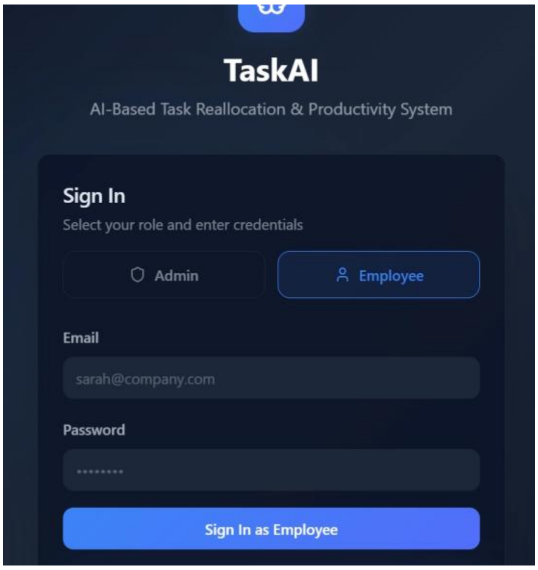
- The system uses a **rule-based AI suitability scoring algorithm** to assign tasks to employees in a fair and efficient manner.
- Each employee is evaluated based on skill match, current workload, and availability before assigning a task.

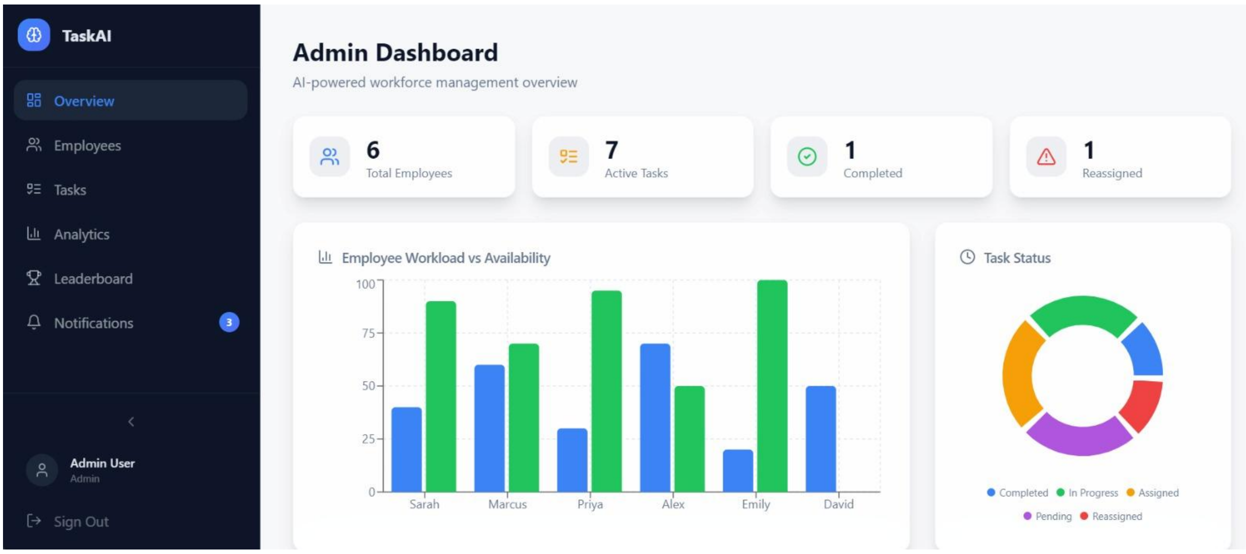
- A suitability score is calculated for every employee-task pair using predefined weighted factors.
- The employee with the highest suitability score is selected using a greedy selection approach.
- When an employee is absent or on leave, the system automatically reallocates pending tasks to available employees.
- This algorithm ensures continuous workflow, balanced workload, and optimal resource utilization.

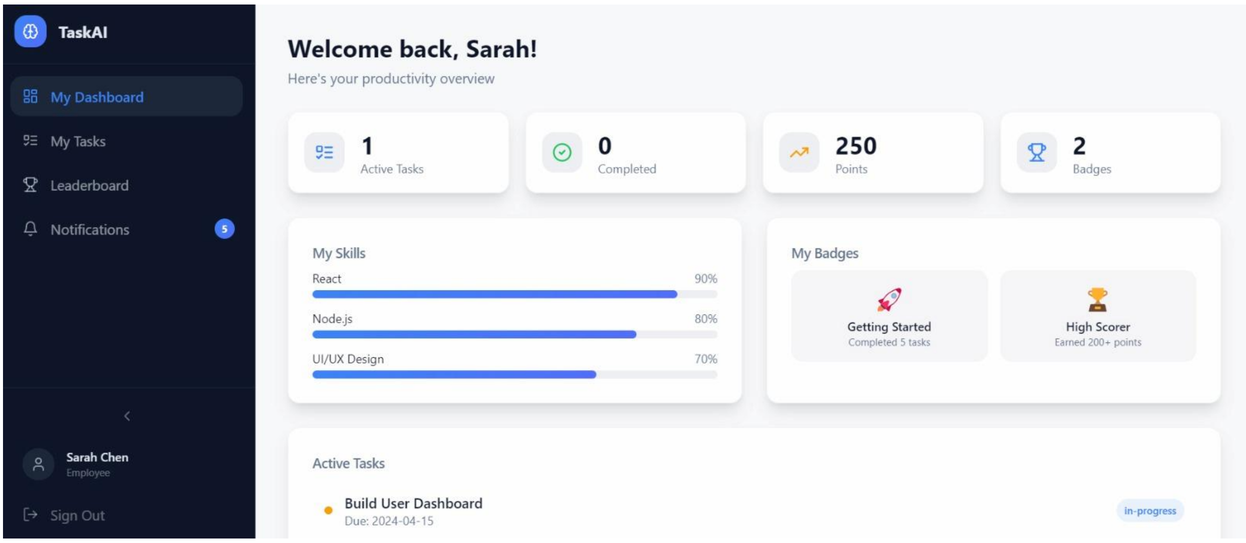
Algorithm 2: Gamification & Performance Analytics Algorithm

- A **reward-based scoring algorithm** is used to improve employee motivation and engagement.
- Employees earn points for timely task completion, which contribute to badges and leaderboard rankings.
- Gamification promotes accountability, healthy competition, and consistent performance.
- Performance analytics use aggregation algorithms to collect workload and task completion data.
- The system analyzes completed, pending, and reassigned tasks in real time.
- Interactive dashboards help management make data-driven decisions to improve productivity.

Project Screenshots:




TaskAI
Overview
Employees
Tasks
Analytics
Leaderboard
Notifications 3
Admin User
Admin
Sign Out
Admin Dashboard
AI-powered workforce management overview
6 Total Employees
7 Active Tasks
1 Completed
1 Reassigned
Employee Workload vs Availability
100
75
50
25
0
Sarah
Marcus
Priya
Alex
Emily
David
Task Status
Completed
In Progress
Assigned
Pending
Reassigned


TaskAI
My Dashboard
My Tasks
Leaderboard
Notifications 5
Sarah Chen
Employee
Sign Out
Welcome back, Sarah!
Here's your productivity overview
1 Active Tasks
0 Completed
250 Points
2 Badges
My Skills
React 90%
Node.js 80%
UI/UX Design 70%
My Badges
Getting Started
Completed 5 tasks
High Scorer
Earned 200+ points
Active Tasks
Build User Dashboard
Due: 2024-04-15
in-progress

State of Publication:

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